Supervyse Tool User Guide

Revision 1.0

11/2024

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1 Introduction

This document is aimed at providing guideline to use Supervyse Tools.

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Revision History

Revision Number	Date	Author	Reviewer	Summary
1.0	2023-07-05	-	-	OPF RV23.06.1 Release Document

2 Insyde SPV IPMI User Guide

2.1 Introduction

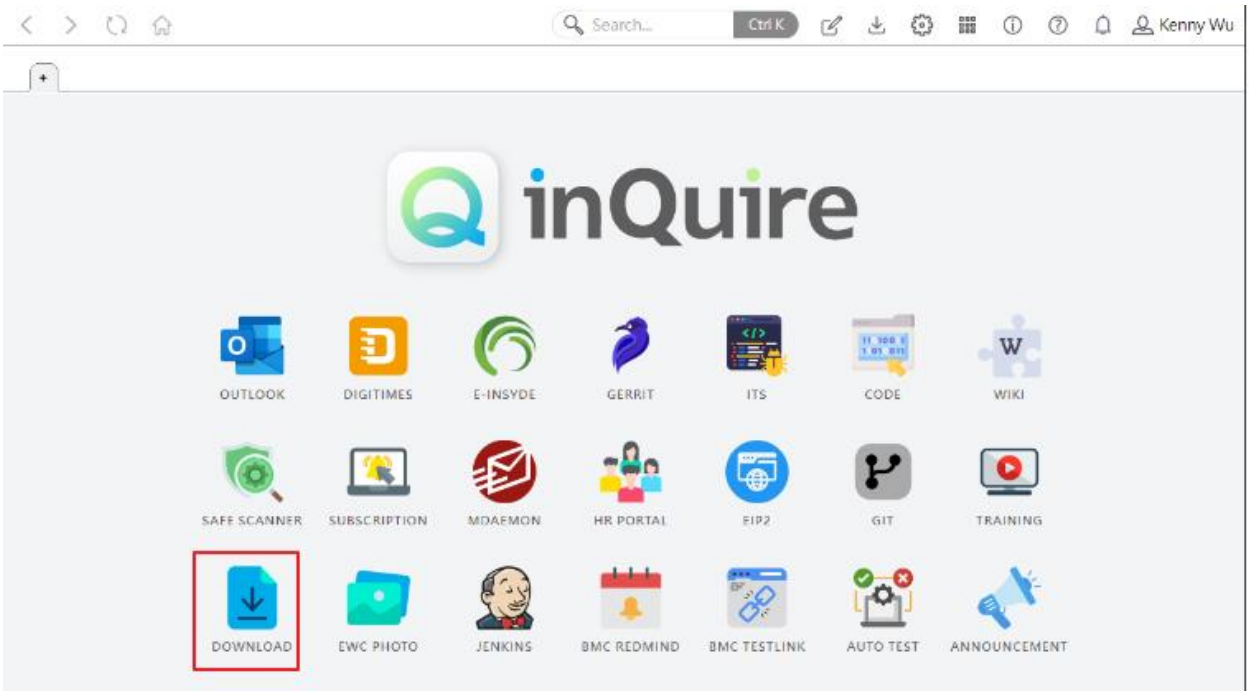
- SPV_IPMI is the Insyde Supervyse ipmitool, which is based on ipmitool(An open-source tool for controlling IPMI-enabled systems).
- This document introduces how to use the Insyde customized ipmitool to communicate with Supervyse OPF BMC more easily and effectively.

2.2 Download

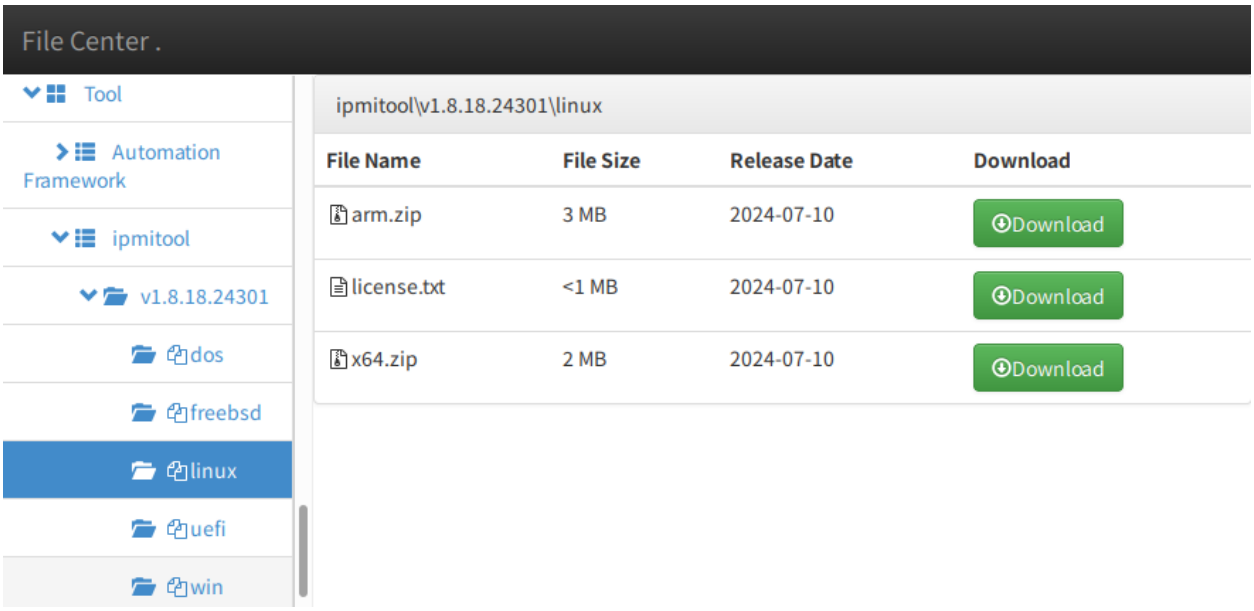
You can download files through the Inquire2 tool provided by Insyde.
InQuire2 is a program that provides easy, centralized access to all of Insyde Software's online cloud services.
We can find Tools using the Download service, follow the path Tool > ipmitool > <verison> > OS.

```
I.e. Tool > ipmitool > v1.8.18.24301 > linux > x64.zip
```

At first, click Download first



And then go to Tool > ipmitool > linux (OS), and click download button at right screen.



2.3 Insyde IPMITool

2.4 Introduction

The Insyde customized ipmitool **SPV_IPMI** usage is very similar to ipmitool.
The only difference is extra Insyde utilities are provided by the "isc" command.

2.4.1 Execution File

We provide a variety of execution programs in different operating systems.
The extension of the program running on different operating systems will be different.

	Windows	Linux	FreeBSD	DOS	UEFI
File Name	spv_ipmi.exe	spv_ipmi	spv_ipmi	spv_ipmi.exe	spv_ipmi.efi

NOTE:
1. Please confirm that you have permission to execute on **Linux** or **FreeBSD**.

```
# Make user have execute permission.
# chmod - change file modes.
#   Parameter:
#        x - Set execute bits to enable.
#
$ chmod +x spv_ipmi
```

2. A dot slash is a dot followed immediately by a forward slash (./).
It is used in **Linux** or **FreeBSD** to execute a program in the current directory.

```
# Execute a program and show help.
./spv_ipmi -h
```

If you don’t want to enter './' in front of the executable file.
You can modify the .bashrc file set environment variables 'PATH' and re-login.

```
# .bashrc file
$ gedit ~/.bashrc

# Add a line of command to the file '.bashrc' set environment variables.
export PATH=$PATH:./
```

2.4.2 Example

We provide an example: Pseudo command line

```
# Show help
./spv_ipmi -h
```

In our example, we show that **$** means the line is an input command.
The beginning of the **#** symbol represents a comment.
You can specify the interface you want to use.
spv_ipmi Please replace the executable file belonging to your current operating system by yourself.

There will be different input methods in different operating systems:
The command prompt displayed will also be different.
* Windows - If the executable file is on the D drive

```
# Show help
D:\>spv_ipmi.exe -h
```

- Dos - If booting via USB and the file is in the root folder.

```
# Show help
C:\>spv_ipmi.exe -h
```

- Linux - If the file is in the home directory.

```
# Show help
$ cd ~
./spv_ipmi -h
```

- FreeBSD - If the file is in the home directory.

```
# Show help
$ cd ~
./spv_ipmi -h
```

- UEFI Boot to USB file system(fs0) under UEFI SHELL.

```
# Show help
fs0:\>spv_ipmi.efi -h
```

2.4.3 Version

```
./spv_ipmi isc version

Insyde Utility Tool 1.8.18.24301
```

1.8.18 is ipmitool version and 24031 is Insyde SPV_IPMI version, for last part, 24301, 24 represents 2024, 3 means Q3, and 1 is the first week Of Q3.

2.5 Interfaces

- SPV_IPMI tool supported interfaces are visible in the usage help output.

- Except LAN and LAN plus belong to remote solution(out-of-band), the others local solution(in-band).

- Commonly used interfaces are listed here. For details, please refer to ipmitool readme or IPMI SPEC v2.0.

2.5.1 Support

Supported interfaces

	Windows	Linux	FreeBSD	DOS	UEFI
lan	v	v	v	-	-
lanplus	v	v	v	-	-
ms	v	-	-	-	-
imb	v	v	-	-	-
open	-	v	v	-	-
kcs	-	-	-	v	v
h2o	-	-	-	-	v

2.5.2 Description & Usage

2.5.2.1 lan

- The lan interface communicates with the BMC over an Ethernet LAN connection using UDP(User Datagram Protocol) under IPv4 or IPv6.
  UDP datagrams are formatted to contain IPMI request/response messages with a IPMI session headers and RMCP(Remote Management Control Protocol) headers.

```
./spv_ipmi -I lan -H 192.168.59.88 -U root -P 0penBmc
```

2.5.2.2 lanplus

- **Like the lan interface**, the lanplus interface communicates with the BMC over an Ethernet LAN connection using UDP under IPv4 or IPv6.
  The difference is that the lanplus interface uses the **RMCP+ protocol** as described in the IPMI v2.0 specification.
  RMCP+ allows for improved authentication and data integrity checks, as well as encryption and the ability to carry multiple types of payloads.

- Due to the fact that we are based on ipmitool v1.8.18, adjustments need to be made to the cipher suite in order to communicate with OpenBMC via the network.
  In OpenBmc, the default value for the cipher suite is 17.

```
./spv_ipmi -I lanplus -H 192.168.59.88 -U root -P 0penBmc -C 17
```

2.5.2.3 ms

- The interface 'ms' is use Microsoft Windows built-in driver **Microsoft IPMI driver (ipmidrv.sys)**
  The Microsoft IPMI driver and WMI IPMI provider supply data from BMC operations to the operating system.
  https://docs.microsoft.com/en-us/previous-versions/windows/desktop/ipmiprv/ipmi-provider
  The IPMI driver is a WDM kernel mode driver that communicates with the BMC using KCS (Keyboard Controller Style).

  NOTE:
  1. Only support MS Windows.
  2. The driver will be automatically generated when the platform you are using supports IPMI.

```
./spv_ipmi -I ms
```

2.5.2.4 imb

- The imb interface supports the Intel IMB (Intel Inter-module Bus) Interface through the /dev/imb device.
  Windows : Intel IMB interface is a KCS(Keyboard Controller Style) interface for Windows.
  You can download Intel IMB driver

```
./spv_ipmi -I imb
```

2.5.2.5 kcs

- The KCS is Connect the BMC and SMI(System Management Interrupt) handler, and transfer SMS(System Management Software) messages.
  Only support DOS & UEFI.

```
./spv_ipmi -I kcs
```

2.5.2.6 h2o

- The h2o interface is based on the Insyde BIOS. It mainly communicates with BMC through the protocol provided by Insyde BIOS.
  Only support UEFI.

```
./spv_ipmi -I h2o
```

2.5.2.7 Open

- The open interface utilizes the OpenIPMI kernel device driver.
- Since we use the parameter **'sudo'** when loading the kernel driver, we also need to add the parameter **'sudo'** when using SPV_IPMI communicate with BMC through the interface OpenIpmi(KCS).
- Please make sure it has been **loaded IPMI driver**.

```
$ sudo spv_ipmi -I open
```

2.5.2.7.1 How to Load OpenIPMI Driver

- Linux: Using Ubuntu as an example

```
 # This provides a userland IOCTL interface for the IPMI driver, each open file for
this device ties in to the message handler as an IPMI user.
 $ sudo modprobe ipmi_devintf

 # A driver can support KCS interfaces.
 $ sudo modprobe ipmi_si
```

- FreeBSD:

```
# Load a IPMI driver into the kernel.
 $ sudo kldload ipmi

 # Display status of dynamic kernel linker.
 $ sudo kldstat
```

2.5.3 Default

- If no interface is specified, it will use KCS by default.
  - There are different interfaces for In-band(local solution) KCS transmission in different OS.
- The tool will specify the default KCS interface according to different OS.
  - Linux/FreeBSD - Open IPMI interface.
  - Windows - MS interface(Microsoft IPMI driver).
  - DOS/UEFI - KCS interface.

2.6 Systems Management

The management of server systems and network equipment can be roughly divided into two methods: In-Band (IB) and Out-of-Band (OOB).

2.6.1 In-Band

- The in-band management through KCS is based on in-band connectivity and software that must be installed on the remote system being managed and only works after the operating system has been booted.
- Prerequisite: The manager must be in the system of the managed device. HOST must be booted to OS first.
- No specify interface is mean use KCS interface(Different OS will be different KCS interface).
    - For details, please see the chapter Default
- When you use the UEFI version of the tool running on Insyde BIOS, you may consider switching to the h2o interface. This interface will utilize the default in-band management provided by BIOS to communicate with BMC.
    - Example: In an environment without a provided KCS interface, we can switch to using the H2O interface. In this case, the default communication between BIOS and BMC is through SSIF. Therefore, the underlying communication of IPMItool is done through SSIF and BMC.
    - Currently, it has been confirmed that the platforms provided by NVIDIA, Starship and Falcon, do not have KCS functionality. Please specify the use of the h2o interface.

```
./spv_ipmi -I h2o isc
```

- Usage: spv_ipmi [options...] <command>

```
./spv_ipmi isc

Insyde Utility Tool 1.8.18.22202

isc commands:
  bmc        - BMC firmware update
  bios       - BIOS firmware update
  pld        - PLD firmware update
  hpm        - Update HPM components using PICMG HPM.1 file
  modmac     - Modify MAC address
  chkalert   - PEF alert auto test
  sdr        - Sensor Data Record
  fru        - Field Replaceable Unit
  spd        - Serial Presence Detect
  mntiso     - Mount Web CD-ROM image
  version    - Tool version
```

2.6.2 Out-of-Band

- The out-of-band management involves the use of remote management interfaces for managing and networking equipment.
- Prerequisite: The network channel must be normal and the network can be connected.
- It is recommended to use lanplus - IPMI v2.0 RMCP+ LAN Interface to operate, because the Remote Management and Control Protocol+ (RMCP+) is encryption network protocol.
- Usage: spv_ipmi [options...]
- Common use options

- -I intf Interface to use
- -H hostname Remote host name for **LAN** or **LANPLUS** interface
- -U username Remote session username
- -P password Remote session password
- -C ID Cipher suite ID (used by lanplus interface)

- Support Commands

spv_ipmi -I <intf> -H <hostname> -U <username> -P <password> <cmd>

```
./spv_ipmi -I lanplus -H 192.168.59.88 -U root -P OpenBmc -C 17 isc

Insyde Utility Tool 1.8.18.22202

isc commands:
  bmc        - BMC firmware update
  bios       - BIOS firmware update
  pld        - PLD firmware update
  hpm        - Update HPM components using PICMG HPM.1 file
  modmac     - Modify MAC address
  chkalert   - PEF alert auto test
  sdr        - Sensor Data Record
  fru        - Field Replaceable Unit
  spd        - Serial Presence Detect
  mntiso     - Mount Web CD-ROM image
  version    - Tool version
```

2.7 IPMItool Commands

- IPMItool provides a variety of commands to communicate with BMC, we only list examples of FW update commands here.

2.7.1 HPM FW Update

- For HPM update commands, there is no distinction for FW because the HPM file will describe which type it currently belongs to.

2.7.1.1 BMC FW Update

- Following example shows using the IPMItool command "hpm" to upload the BMC image file to BMC 192.168.11.144. With large buffer 30000 bytes.

```
./spv_ipmi -I lanplus -U root -P 0penBmc -H 192.168.11.144 -C 17 -N 10 -z 30000 hpm
upgrade bmc-image.hpm force activate
```

- Parameter description
    - -I lanplus: Specifies the lanplus communication interface, which supports IPMI v2.0 and enables encryption.
    - -U root: Sets the username for IPMI login, here 'root'.
    - -P 0penBmc: Sets the password for IPMI login, here '0penBmc'.
    - -H 192.168.11.144: Defines the target IP address of the IPMI-enabled device.
    - -C 17: Sets the cipher suite ID for encryption, with 17 being the selected suite. This option is specific to lanplus.
    - -N 10: Specifies a timeout of 10 seconds for the IPMI command, only available with the lanplus interface.
    - -z 30000: Sets the maximum packet size in bytes for data transmission, here 30000, and is only usable with lanplus.

```
insyde@insyde-VirtualBox:/media/sf_Work/Work_Git/Supervyse_IPMI/build/linux/x64_release$
 ./spv_ipmi -I lanplus -U root -P 0penBmc -H 192.168.11.13 -C 17 -N 10 -z 30000 hpm upgr
ade image-update-ast2600.hpm force activate
Setting large buffer to 30000

PICMG HPM.1 Upgrade Agent 1.0.9:

Validating firmware image integrity...OK
Performing preparation stage...
    Invalid image file for product 2817

    Image Information
        Device Id : 0x22
        Prod   Id : 0x0003
        Manuf  Id : 0x00b009
    Board Information
        Device Id : 0x22
        Prod   Id : 0x0b01
        Manuf  Id : 0x00b009
 Continue ignoring DeviceID/ProductID/ManufacturingID (Y/N): y

Services may be affected during upgrade. Do you wish to continue? (y/n): y
OK

Performing upgrade stage:

--------------------------------------------------------------------------------
|ID  | Name        |                     Versions                          | %  |
|    |             |      Active     |      Backup     |      File       |    |
|----|-------------|-----------------|-----------------|-----------------|----|
|*  0|BMC          |   0.ff 0706C0A4 | ---.-- -------- |  32.11 00000000 |100%|
|    |Upload Time: 01:41             | Image Size: 57333760 bytes                |
--------------------------------------------------------------------------------
(*) Component requires Payload Cold Reset
Performing activation stage:
    Waiting firmware activation...OK

Firmware upgrade procedure successful
```

2.7.1.2 BIOS FW Update

- Following example shows how to use the IPMItool command "hpm" to upload the BIOS image file ArcherCity_0046.hpm. With force activate and large buffer 30000 configured.

```
./spv_ipmi -I lanplus -U root -P 0penBmc -H 192.168.10.68 -C 17 -z 30000 hpm upgrade
ArcherCity_0046.hpm  force activate
```

- Parameter description
    - -I lanplus: Specifies the lanplus communication interface, which supports IPMI v2.0 and enables encryption.
    - -U root: Sets the username for IPMI login, here 'root'.
    - -P 0penBmc: Sets the password for IPMI login, here '0penBmc'.
    - -H 192.168.10.68: Defines the target IP address of the IPMI-enabled device.
    - -C 17: Sets the cipher suite ID for encryption, with 17 being the selected suite. This option is specific to lanplus.
    - -N 10: Specifies a timeout of 10 seconds for the IPMI command, only available with the lanplus interface.
    - -z 30000: Sets the maximum packet size in bytes for data transmission, here 30000, and is only usable with lanplus.

```
insyde@insyde-VirtualBox:/media/sf_Work/Work_Git/Supervyse_IPMI_BIN/linux/x64$
./spv_ipmi -I lanplus -U root -P OpenBmc -H 192.168.10.68 -C 17 hpm upgrade Arc
herCity_0046.hpm -z 30000 force activate
Setting large buffer to 30000

PICMG HPM.1 Upgrade Agent 1.0.9:

Validating firmware image integrity...OK
Performing preparation stage...
    Invalid device ID 22

    Image Information
        Device Id : 0x20
        Prod   Id : 0x3558
        Manuf  Id : 0x001c4c
    Board Information
        Device Id : 0x22
        Prod   Id : 0x0100
        Manuf  Id : 0x00b009
 Continue ignoring DeviceID/ProductID/ManufacturingID (Y/N): y

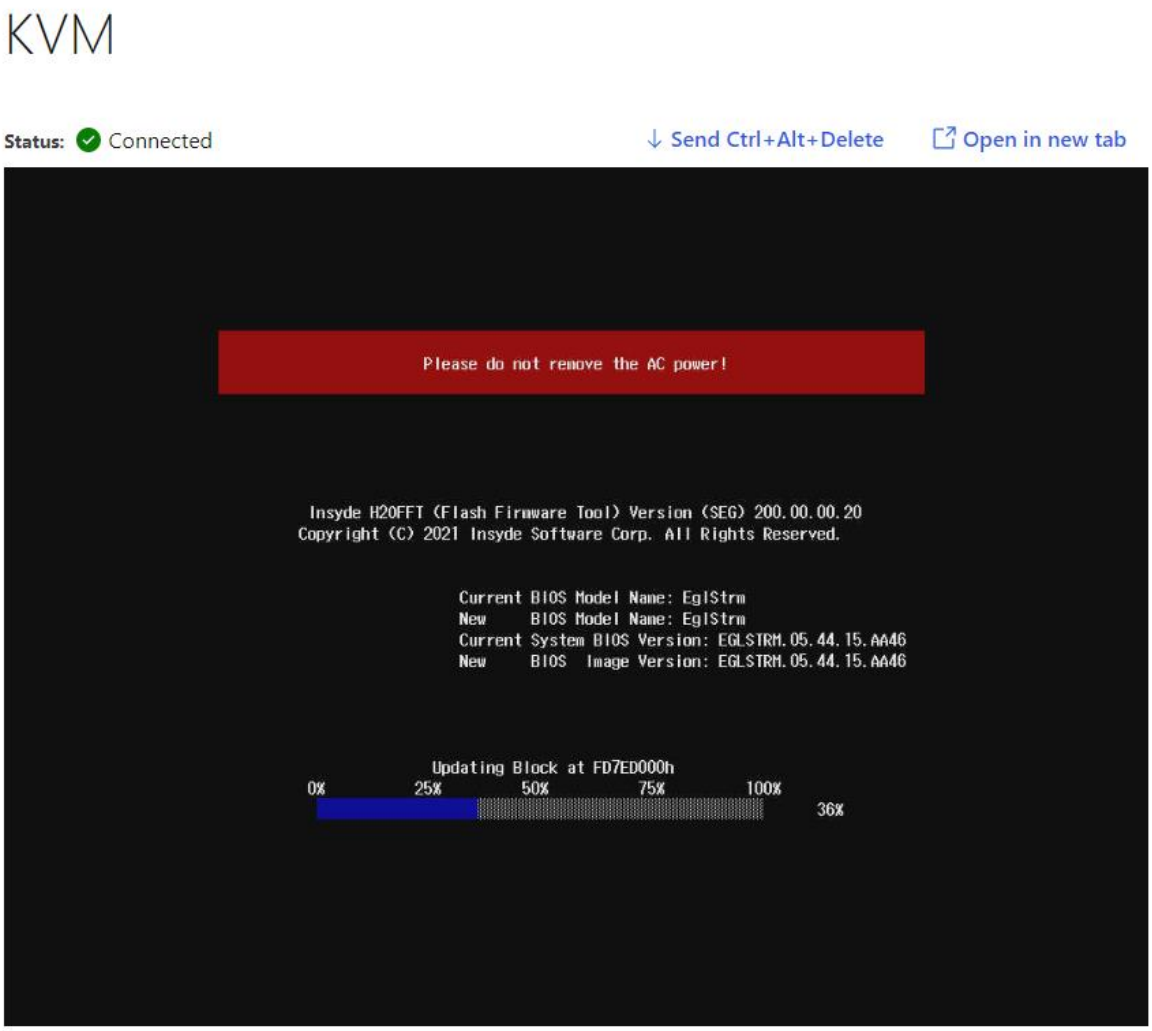
Services may be affected during upgrade. Do you wish to continue? (y/n): y
OK

Performing upgrade stage:

-------------------------------------------------------------------------------
|ID  | Name        |                     Versions                        | %  |
|    |             |      Active     |      Backup     |      File       |    |
|----|-------------|-----------------|-----------------|-----------------|----|
|*  2|BIOS         |   5.00 00000000 | ---.-- -------- |   3.52 00170013 100%||
|    |Upload Time: 00:27             | Image Size: 17769376 bytes
|
-------------------------------------------------------------------------------
(*) Component requires Payload Cold Reset
Performing activation stage:
    Waiting firmware activation...OK

Firmware upgrade procedure successful
```

- After execution, it will not update immediately. It will create a virtual device with a BIOS



2.8 Insyde Commands

- The following is an introduction to using Insyde OEM Command and OPF for communication.
- The examples provided are all about communicating with BMC through remote access on Ubuntu. If you want to switch to In-Band, please refer to the previous instructions.

2.8.1 FW Update

2.8.1.1 BMC FW Update

2.8.1.1.1 Get BMC Version

- Get the version number of BMC.

```
./spv_ipmi -I lanplus -U root -P OpenBmc -H 192.168.10.68 -C 17 isc bmc version
```

- Parameter description
    - -I lanplus: Specifies the lanplus communication interface, which supports IPMI v2.0 and enables encryption.
    - -U root: Sets the username for IPMI login, here 'root'.
    - -P 0penBmc: Sets the password for IPMI login, here '0penBmc'.
    - -H 192.168.10.68: Defines the target IP address of the IPMI-enabled device.
    - -C 17: Sets the cipher suite ID for encryption, with 17 being the selected suite. This option is specific to lanplus.

```
insyde@insyde-VirtualBox:/media/sf_Work/Work_Git/Supervyse_IPMI_BIN/linux/x64$ .
/spv_ipmi -I lanplus -U root -P 0penBmc -H 192.168.10.68 -C 17 isc bmc version
Type          : BMC Active Image
Version       : 35.1.0
```

2.8.1.1.2 Get BMC Status

- Only the idle state allows updates, so you need to first check the current BMC status.

```
./spv_ipmi -I lanplus -U root -P 0penBmc -H 192.168.11.144 -C 17 isc bmc status
```

- Parameter description
    - -I lanplus: Specifies the lanplus communication interface, which supports IPMI v2.0 and enables encryption.
    - U root: Sets the username for IPMI login, here 'root'.
    - -P 0penBmc: Sets the password for IPMI login, here '0penBmc'.
    - -H 192.168.11.144: Defines the target IP address of the IPMI-enabled device.
    - -C 17: Sets the cipher suite ID for encryption, with 17 being the selected suite. This option is specific to lanplus.

```
insyde@insyde-VirtualBox:/media/sf_Work/Work_Git/Supervyse_IPMI_BIN/linux/x64$ .
/spv_ipmi -I lanplus -U root -P 0penBmc -H 192.168.11.144 -C 17 isc bmc status
Status                : Idle
Percentage            : 0
Hash Check            : None requested
```

2.8.1.1.3 Reset BMC Update Mode

- Set the BMC update mode to Idle.

```
./spv_ipmi -I lanplus -U root -P 0penBmc -H 192.168.11.144 -C 17 isc bmc reset
```

- Parameter description
    - -I lanplus: Specifies the lanplus communication interface, which supports IPMI v2.0 and enables encryption.
    - -U root: Sets the username for IPMI login, here 'root'.
    - -P 0penBmc: Sets the password for IPMI login, here '0penBmc'.
    - -H 192.168.11.144: Defines the target IP address of the IPMI-enabled device.
    - -C 17: Sets the cipher suite ID for encryption, with 17 being the selected suite. This option is specific to lanplus.

```
insyde@insyde-VirtualBox:/media/sf_Work/Work_Git/Supervyse_IPMI_BIN/linux/x64$ .
/spv_ipmi -I lanplus -U root -P 0penBmc -H 192.168.11.144 -C 17 isc bmc reset
Reset BMC update mode to IDLE.
```

2.8.1.1.4 Get BMC SDK Version

OPF does not yet support.

2.8.1.1.5 Get BMC SDK Name

OPF does not yet support.

2.8.1.1.6 BMC FW Update

- Use the HPM file to update the BMC Image through Insyde Command.

```
./spv_ipmi -I lanplus -U root -P 0penBmc -H 192.168.11.144 -C 17 isc bmc update bmc-image.hpm
```

- Parameter description
  - -I lanplus: Specifies the lanplus communication interface, which supports IPMI v2.0 and enables encryption.
  - -U root: Sets the username for IPMI login, here 'root'.
  - -P 0penBmc: Sets the password for IPMI login, here '0penBmc'.
  - -H 192.168.11.144: Defines the target IP address of the IPMI-enabled device.
  - -C 17: Sets the cipher suite ID for encryption, with 17 being the selected suite. This option is specific to lanplus.

```
insyde@insyde-VirtualBox:/media/sf_Work/Work_Git/Supervyse_IPMI_BIN/linux/x64$ ./spv_ipmi -I lanplus -U root -P 0penBmc -H 192.168.11.144 -C 17 isc bmc update bmc-hpm.bin
Firmware upload start.
Image size      : 33.05 (mb)
100 % [########################################]
Firmware upload complete.

Get firmware update key.
Enter firmware update mode.
Set file name.

Status: Programming Image (0x04)
100 % [########################################]
Status: Ready - Successfully Update (0x05)

Ready for reboot.
```

2.8.1.1.7 BMC FW Update via USB

- Due to the slow speed caused by file transmission through In-Band updates, we will utilize virtual devices to accelerate the file transfer process.

2.8.1.1.8 Backup / Restore User Configuration File

- Example: Backup user configuration File to 'backup.bin'.
- ```
  ./spv_ipmi -I lanplus -U root -P 0penBmc -C 17 -H 192.168.10.102 isc bmc cfg backup backup.bin
  ```

- Parameter description

  - -I lanplus: Specifies the lanplus communication interface, which supports IPMI v2.0 and enables encryption.
  - -U root: Sets the username for IPMI login, here 'root'.
  - -P 0penBmc: Sets the password for IPMI login, here '0penBmc'.
  - -H 192.168.10.102: Defines the target IP address of the IPMI-enabled device.
  - -C 17: Sets the cipher suite ID for encryption, with 17 being the selected suite. This option is specific to lanplus.

```
insyde@insyde-VirtualBox:/media/sf_Work/Work_Git/Supervyse_IPMI_BIN/linux/x64$ .
/spv_ipmi -I lanplus -U root -P 0penBmc -C 17 -H 192.168.10.102 isc bmc cfg back
up backup.bin
Backup user configuration file.
Please wait for the process.

Download Size: 17 KB
100 % [########################################]

Success to download configuration file.
```

- Example: Restore user configuration from file 'backup.bin'.

  - The restore must wait for the BMC to restart to take effect.

```
 ./spv_ipmi -I lanplus -U root -P 0penBmc -C 17 -H 192.168.10.102 isc bmc cfg restore
backup.bin
```

- Parameter description

  - -I lanplus: Specifies the lanplus communication interface, which supports IPMI v2.0 and enables encryption.
  - -U root: Sets the username for IPMI login, here 'root'.
  - -P 0penBmc: Sets the password for IPMI login, here '0penBmc'.
  - -H 192.168.10.102: Defines the target IP address of the IPMI-enabled device.
  - -C 17: Sets the cipher suite ID for encryption, with 17 being the selected suite. This option is specific to lanplus.
```

```
insyde@insyde-VirtualBox:/media/sf_Work/Work_Git/Supervyse_IPMI_BIN/linux/x64$
/spv_ipmi -I lanplus -U root -P 0penBmc -C 17 -H 192.168.10.102 isc bmc cfg res
ore  backup.bin

Note: BMC will automatically restart after the restoration is successful!! (Y/N
:
y
Start upload user configuration file.
Transfer user configuration file.
100 % [########################################]
Finish upload user configuration file.
Verification : Pass

Restore user configuration file.
Please wait for the process.
Restart BMC.

Success to restore user configuration file.
```

- Example: Force restore user configuration from file 'backup.bin'.
    - BMC will be restarted immediately after restore.

```
./spv_ipmi -I lanplus -U root -P 0penBmc -C 17 -H 192.168.10.102 isc bmc cfg restore backup.bin force
```

- Parameter description
    - -I lanplus: Specifies the lanplus communication interface, which supports IPMI v2.0 and enables encryption.
    - -U root: Sets the username for IPMI login, here 'root'.
    - -P 0penBmc: Sets the password for IPMI login, here '0penBmc'.
    - -H 192.168.10.102: Defines the target IP address of the IPMI-enabled device.
    - -C 17: Sets the cipher suite ID for encryption, with 17 being the selected suite. This option is specific to lanplus.

```
insyde@insyde-VirtualBox:/media/sf_Work/Work_Git/Supervyse_IPMI_BIN/linux/x64$ ./spv_ipmi -I lanplus -U root -P 0penBmc -C 17 -H 192.168.10.102 isc bmc cfg restore  backup.bin force
Start upload user configuration file.
Transfer user configuration file.
100 % [########################################]
Finish upload user configuration file.
Verification : Pass

Restore user configuration file.
Please wait for the process.
Restart BMC.

Success to restore user configuration file.
```

2.8.1.1.9 Export Download Diagnostic Log

2.8.1.2 BIOS FW Update

- Currently, only updating the BIOS through virtual devices is supported.

2.8.1.2.1 Get BIOS status

- BIOS updates are performed through virtual device updates. The tool simply transfers the file to the BMC, which mounts a virtual device containing the USB. The update will be automatically applied when the host restarts.
- We can obtain the current mounting information of virtual devices by acquiring their status.
- Only the idle state allows updates, so you need to first check the current BIOS update status.

```
./spv_ipmi -I lanplus -U root -P 0penBmc -H 192.168.11.144 -C 17 isc bios status
```

- Parameter description
    - -I lanplus: Specifies the lanplus communication interface, which supports IPMI v2.0 and enables encryption.

- -U root: Sets the username for IPMI login, here 'root'.
- -P 0penBmc: Sets the password for IPMI login, here '0penBmc'.
- -H 192.168.11.144: Defines the target IP address of the IPMI-enabled device.
- -C 17: Sets the cipher suite ID for encryption, with 17 being the selected suite. This option is specific to lanplus.

```
insyde@insyde-VirtualBox:/media/sf_Work/Work_Git/Supervyse_IPMI_BIN/linux/x64$ .
/spv_ipmi -I lanplus -U root -P 0penBmc -H 192.168.11.144 -C 17 isc bios status
BIOS Update Method    : Virtual Media
BIOS update mode      : No
Image uploaded        : No
Virtual USB ready     : No
```

2.8.1.2.2 Reset BIOS Update Mode

```
./spv_ipmi -I lanplus -U root -P 0penBmc -H 192.168.11.144 -C 17 isc bios reset
```

- Parameter description
  - -I lanplus: Specifies the lanplus communication interface, which supports IPMI v2.0 and enables encryption.
  - -U root: Sets the username for IPMI login, here 'root'.
  - -P 0penBmc: Sets the password for IPMI login, here '0penBmc'.
  - -H 192.168.11.144: Defines the target IP address of the IPMI-enabled device.
  - -C 17: Sets the cipher suite ID for encryption, with 17 being the selected suite. This option is specific to lanplus.

```
insyde@insyde-VirtualBox:/media/sf_Work/Work_Git/Supervyse_IPMI_BIN/linux/x64$ .
/spv_ipmi -I lanplus -U root -P 0penBmc -H 192.168.11.144 -C 17 isc bios reset

Current is exit BIOS firmware update mode.
```

2.8.1.2.3 Update BIOS Firmware

- You need to use the HPM image.

```
 ./spv_ipmi -I lanplus -U root -P 0penBmc -H 192.168.11.144 -C 17 isc bios update
ArcherCity_0046.hpm
```

- Parameter description
  - -I lanplus: Specifies the lanplus communication interface, which supports IPMI v2.0 and enables encryption.
  - -U root: Sets the username for IPMI login, here 'root'.
  - -P 0penBmc: Sets the password for IPMI login, here '0penBmc'.
  - -H 192.168.11.144: Defines the target IP address of the IPMI-enabled device.
  - -C 17: Sets the cipher suite ID for encryption, with 17 being the selected suite. This option is specific to lanplus.

```
insyde@insyde-VirtualBox:/media/sf_Work/Work_Git/Supervyse_IPMI_BIN/linux/x64$ .
/spv_ipmi -I lanplus -U root -P OpenBmc -H 192.168.10.68 -C 17 isc bios update A
rcherCity_0046.hpm

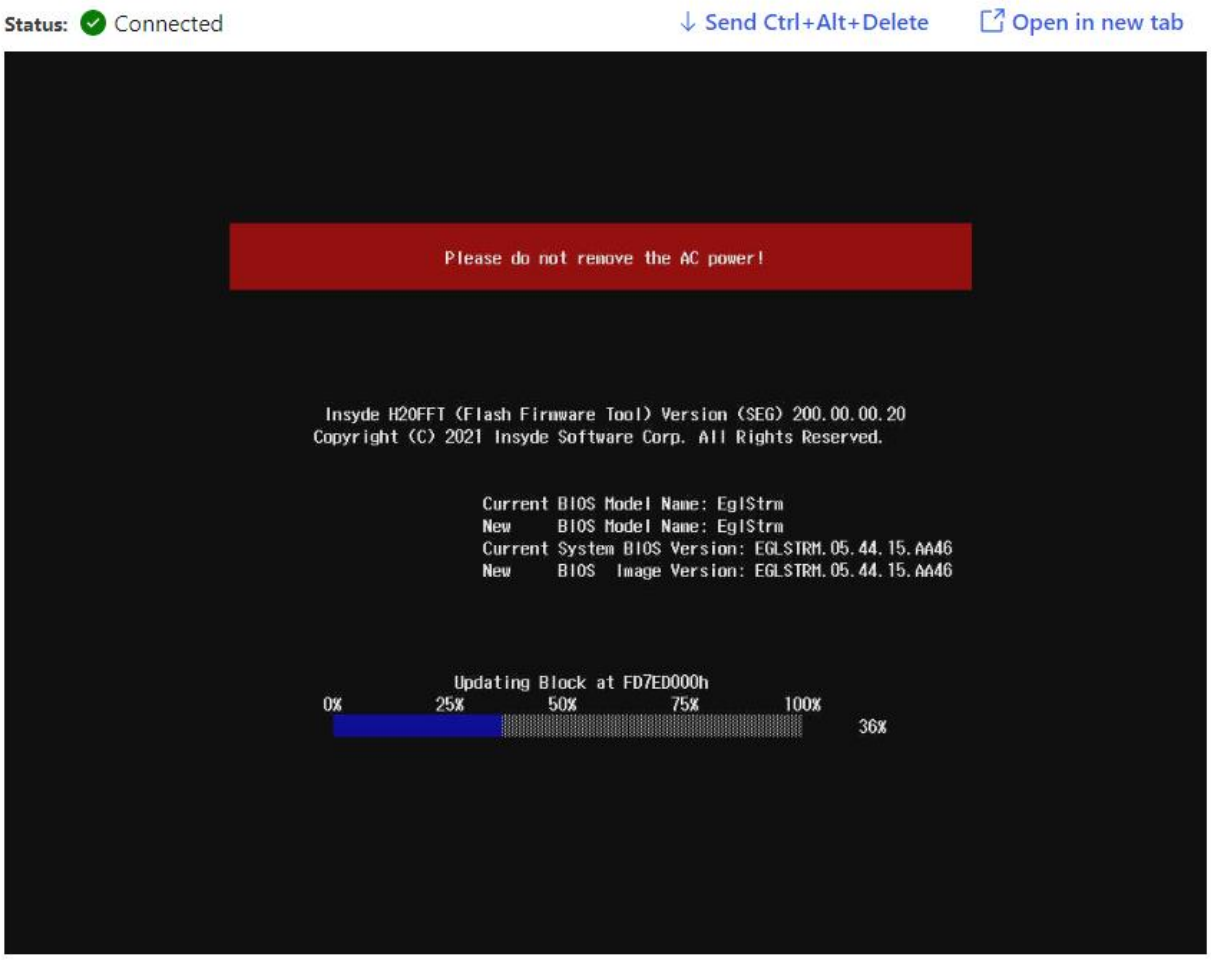
Current is exit BIOS firmware update mode.

Firmware upload start.
Image size      : 16.95 (mb)
100 % [########################################]
Firmware upload complete.

Enter BIOS update mode
Waiting Virtual USB ready for BIOS to start FW update...
Virtual USB is ready
BIOS will be updated until reboot next time!
```

- After execution, it will not update immediately. It will create a virtual device with a BIOS Image inside. Restarting the host will allow for BIOS updates.

KVM



2.8.1.2.4 Clear CMOS

- Clear CMOS is only implemented on specific platforms and is related to hardware configuration.

```
./spv_ipmi -I lanplus -U root -P 0penBmc -H 192.168.11.144 -C 17 isc bios ccmos
```

- Parameter description
    - -I lanplus: Specifies the lanplus communication interface, which supports IPMI v2.0 and enables encryption.
    - -U root: Sets the username for IPMI login, here 'root'.
    - -P 0penBmc: Sets the password for IPMI login, here '0penBmc'.
    - -H 192.168.11.144: Defines the target IP address of the IPMI-enabled device.
    - -C 17: Sets the cipher suite ID for encryption, with 17 being the selected suite. This option is specific to lanplus.

```
insyde@insyde-VirtualBox:/media/sf_Work/Work_Git/Supervyse_IPMI_BIN/linux/x64$ .
/spv_ipmi -I lanplus -U root -P 0penBmc -H 192.168.11.144 -C 17 isc bios ccmos

Not support current platform!!

Failed to clear CMOS!!
```

2.8.1.2.5 Download BIOS Firmware

- When the BIOS is updated through SPI, it supports downloading the BIOS FW Image.
- Since our BIOS is updated through a virtual device (not through SPI), it is normal that it cannot be used at the moment.

```
./spv_ipmi -I lanplus -U root -P 0penBmc -H 192.168.11.144 -C 17 isc bios download
```

- Parameter description
    - -I lanplus: Specifies the lanplus communication interface, which supports IPMI v2.0 and enables encryption.
    - -U root: Sets the username for IPMI login, here 'root'.
    - -P 0penBmc: Sets the password for IPMI login, here '0penBmc'.
    - -H 192.168.11.144: Defines the target IP address of the IPMI-enabled device.
    - -C 17: Sets the cipher suite ID for encryption, with 17 being the selected suite. This option is specific to lanplus.

```
insyde@insyde-VirtualBox:/media/sf_Work/Work_Git/Supervyse_IPMI_BIN/linux/x64$ .
/spv_ipmi -I lanplus -U root -P 0penBmc -H 192.168.11.144 -C 17 isc bios downloa
d
Download function only supports SPI mode!!
```

2.8.1.3 PLD FW Update

- OPF does not yet support PLD FW Update

2.8.1.3.1 Get PLD status

- OPF does not yet support.

2.8.1.3.2 Update PLD Firmware

- OPF does not yet support.

2.8.1.3.3 Reset PLD Update Mode

- OPF does not yet support.

2.8.1.3.4 Show PLD Version

- OPF does not yet support.

2.8.2 Other

2.8.2.1 Modify MAC address

- OPF has not been fully verified yet, and MTM (Manufacturing Test Mode) needs to be enabled first.
- NVIDIA platform not yet supported.

2.8.2.2 SDR Update

- Currently, only Intel ArcherCity updates are supported

Not all fields can be updated. Please refer to document **Supervyse OPF SDR** Update **User Guide** for details.

```
./spv_ipmi -I lanplus -U root -P 0penBmc -H 192.168.11.144 -C 17 isc sdr update
FCT.sdr
```

- Parameter description
    - -I lanplus: Specifies the lanplus communication interface, which supports IPMI v2.0 and enables encryption.
    - -U root: Sets the username for IPMI login, here 'root'.
    - -P 0penBmc: Sets the password for IPMI login, here '0penBmc'.
    - -H 192.168.11.144: Defines the target IP address of the IPMI-enabled device.
    - -C 17: Sets the cipher suite ID for encryption, with 17 being the selected suite. This option is specific to lanplus.

```
insyde@insyde-VirtualBox:/media/sf_Work/Work_Git/Supervyse_IPMI_BIN/linux/x64$ .
/spv_ipmi -I lanplus -U root -P 0penBmc -H 192.168.11.144 -C 17 isc sdr update F
CT.sdr

Parsing SDR File: FCT.sdr
Clearing SDR Repository.
SDR successfully erased.
Waiting for write the SDRs to the SDR Repository...

Success to update SDR!!
```

2.8.2.3 FRU Update

- You can update *.fru, but **under OPF, you need to enable MTM (Manufacturing Test Mode) first. Otherwise, even if it shows success, you won't be able to see the correct data**.

```
./spv_ipmi -I lanplus -U root -P 0penBmc -H 192.168.11.144 -C 17 isc fru update 0
FRU.fru
```

- Parameter description
    - -I lanplus: Specifies the lanplus communication interface, which supports IPMI v2.0 and enables encryption.
    - -U root: Sets the username for IPMI login, here 'root'.
    - -P 0penBmc: Sets the password for IPMI login, here '0penBmc'.
    - -H 192.168.11.144: Defines the target IP address of the IPMI-enabled device.
    - -C 17: Sets the cipher suite ID for encryption, with 17 being the selected suite. This option is specific to lanplus.

```
insyde@insyde-VirtualBox:/media/sf_Work/Work_Git/Supervyse_IPMI_BIN/linux/x64$ .
/spv_ipmi -I lanplus -U root -P 0penBmc -H 192.168.11.144 -C 17 isc fru update 0
 FRU.fru
Fru Size          : 512 bytes
Size to Write     : 312 bytes
```

2.8.2.4 Mount ISO

- Insyde provides a way to share ISO with BMC.

2.8.2.4.1 Get VM Mount information

- Get current virtual device information.
    - If you don't know, you can omit the index. You can use "help" to find out how many VMs are available on the current BMC.

```
./spv_ipmi -I lanplus -U root -P 0penBmc -H 192.168.11.144 -C 17 isc mntiso status 1
```

- Parameter description
    - -I lanplus: Specifies the lanplus communication interface, which supports IPMI v2.0 and enables encryption.
    - -U root: Sets the username for IPMI login, here 'root'.
    - -P 0penBmc: Sets the password for IPMI login, here '0penBmc'.
    - -H 192.168.11.144: Defines the target IP address of the IPMI-enabled device.
    - -C 17: Sets the cipher suite ID for encryption, with 17 being the selected suite. This option is specific to lanplus.

```
insyde@insyde-VirtualBox:/media/sf_Work/Work_Git/Supervyse_IPMI_BIN/linux/x64$ .
/spv_ipmi -I lanplus -U root -P 0penBmc -H 192.168.11.144 -C 17 isc mntiso statu
s 1
Media Type         : Unused
Connected Via      : Not connected
Inserted Status    : FALSE
Read Only          : TRUE
```

2.8.2.4.2 Set Web ISO information

- When we load a WEB ISO from a remote server through BMC, we need to mount it.
    - Since the connection process of the lanplus interface already includes the password and account of BMC, there is no need to specify them separately.
    - However, if it comes from the KCS interface, you must specify the password and account of BMC in order to complete the mounting.
- We used a command to BMC to mount the /var/nfsshare/iso/ubuntu-18.04.6-desktop-amd64.iso file from 192.168.11.177 using the NFS protocol.$./spv_ipmi -I lanplus -U root -P 0penBmc -H 192.168.11.144 -C 17 isc mntiso set 1 nfs 192.168.11.177 /var/nfsshare/iso/ubuntu-18.04.6-desktop-amd64.iso ./spv_ipmi -I lanplus -U root -P 0penBmc -H 192.168.11.144 -C 17 isc mntiso set 1 nfs 192.168.11.177 /var/nfsshare/iso/ubuntu-18.04.6-desktop-amd64.iso
- Parameter description
    - -I lanplus: Specifies the lanplus communication interface, which supports IPMI v2.0 and enables encryption.
    - -U root: Sets the username for IPMI login, here 'root'.
    - -P 0penBmc: Sets the password for IPMI login, here '0penBmc'.
    - -H 192.168.11.144: Defines the target IP address of the IPMI-enabled device.
    - -C 17: Sets the cipher suite ID for encryption, with 17 being the selected suite. This option is specific to lanplus.

```
insyde@insyde-VirtualBox:/media/sf_Work/Work_Git/Supervyse_IPMI_BIN/linux/x64$ .
/spv_ipmi -I lanplus -U root -P 0penBmc -H 192.168.11.144 -C 17 isc mntiso set 1
 nfs 192.168.11.177 /var/nfsshare/iso/ubuntu-18.04.6-desktop-amd64.iso
Success to Set share host: '192.168.11.177'
Success to Set path to Image: '/var/nfsshare/iso/ubuntu-18.04.6-desktop-amd64.is
o'
```

2.8.2.4.3 Get Web ISO information

- If we have previously set up WEB ISO, whether through the WEB UI or SPV_IPMI, we can use commands to obtain relevant information.
- For example, let's retrieve information for the first VM.

```
./spv_ipmi -I lanplus -U root -P 0penBmc -H 192.168.11.144 -C 17 isc mntiso get 1
```

- Parameter description
  - -I lanplus: Specifies the lanplus communication interface, which supports IPMI v2.0 and enables encryption.
  - -U root: Sets the username for IPMI login, here 'root'.
  - -P 0penBmc: Sets the password for IPMI login, here '0penBmc'.
  - -H 192.168.11.144: Defines the target IP address of the IPMI-enabled device.
  - -C 17: Sets the cipher suite ID for encryption, with 17 being the selected suite. This option is specific to lanplus.

```
insyde@insyde-VirtualBox:/media/sf_Work/Work_Git/Supervyse_IPMI_BIN/linux/x64$ .
/spv_ipmi -I lanplus -U root -P 0penBmc -H 192.168.11.144 -C 17 isc mntiso get 1
Share Host      : 192.168.11.177
Path To Image   : /var/nfsshare/iso/ubuntu-18.04.6-desktop-amd64.iso
```

2.8.2.4.4 Unmount Web CD-ROM image

- Only when it is mounted can it be properly unmounted.

```
./spv_ipmi -I lanplus -U root -P 0penBmc -H 192.168.11.83 -C 17 isc mntiso umount 1
```

- Parameter description
  - -I lanplus: Specifies the lanplus communication interface, which supports IPMI v2.0 and enables encryption.
  - -U root: Sets the username for IPMI login, here 'root'.
  - -P 0penBmc: Sets the password for IPMI login, here '0penBmc'.
  - -H 192.168.11.83: Defines the target IP address of the IPMI-enabled device.
  - -C 17: Sets the cipher suite ID for encryption, with 17 being the selected suite. This option is specific to lanplus.

```
insyde@insyde-VirtualBox:/media/sf_Work/Work_Git/Supervyse_IPMI_BIN/linux/x64$
./spv_ipmi -I lanplus -U root -P 0penBmc -H 192.168.11.83 -C 17 isc mntiso umou
nt 1
Success to unmount device 1
```

2.8.2.5 SPD

- Detecting memory, this feature is hardware-related and requires hardware support.

```
./spv_ipmi -I lanplus -U root -P 0penBmc -H 192.168.11.144 -C 17 isc spd
```

- Parameter description
    - -I lanplus: Specifies the lanplus communication interface, which supports IPMI v2.0 and enables encryption.
    - -U root: Sets the username for IPMI login, here 'root'.
    - -P 0penBmc: Sets the password for IPMI login, here '0penBmc'.
    - -H 192.168.11.144: Defines the target IP address of the IPMI-enabled device.
    - -C 17: Sets the cipher suite ID for encryption, with 17 being the selected suite. This option is specific to lanplus.

```
insyde@insyde-VirtualBox:/media/sf_Work/Work_Git/Supervyse_IPMI_BIN/linux/x64$ .
/spv_ipmi -I lanplus -U root -P 0penBmc -H 192.168.11.144 -C 17 isc spd
This platform is no supported !
```

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